In Gridworld, we had four possible actions: up, down, left, and right. We call this our action set or actions-space, since we can describe it mathematically as a set, e.g., $A = \{up,down,left,right\}$ where the curly braces indicate a set. (A set in mathematics is just an abstract unordered collection of things with certain operations defined.) So what does it mean to apply a probability distribution over this set of actions?

Probability is actually a very rich and even controversial topic in its own right. There are varying philosophical opinions on exactly what *probability* means. To some people, the probability means that, if you were to flip a coin a very large number of times (ideally an infinite number of times, mathematically speaking) the probability of a fair coin turning up heads is equal to the proportion of heads in that infinitely long sequence of flips. That is, if we flip a fair coin 1,000,000 times, we would expect about half of the flips to be heads and the other half tails, so the probability is equal to that proportion. This is a frequentist interpretation of probability, since probability is interpreted as the long-term frequency of some event repeated many times.

Another school of thought interprets probability only as a degree of belief, a subjective assessment of how much someone can predict an event given the knowledge they currently possess. This degree of belief is often called a *credence*. The probability of a fair coin turning up heads is 0.5 or 50% because, given what we know about the coin, we don't have any reason to predict heads more than tails, or tails more than heads, so we split our belief evenly across the two possible outcomes. Hence, anything that we can't predict deterministically (i.e., with probability 0 or 1, and nothing in between) results from a lack of knowledge.

You're free to interpret probabilities however you want, since it won't affect our calculations, but in this book we tend to implicitly use the credence interpretation of probability. For our purposes, applying a probability distribution over the set of actions in Gridworld, $A = \{up,down,left,right\}$ means we're assigning a degree of belief (a real number between 0 and 1) to each action in the set such that all the probabilities sum to 1. We interpret these probabilities as the probability that an action is the best action to maximize the expected rewards, given that we're in a certain state.